Discovered joint equation

Searched on all Train+Val wafers (Nov 4); this exact f is not what the 5-fold check on the right tests

$$MRR_g = a_g \cdot f + b_g$$

$$f = (\ln P_{ret} - \int P_{ret}) \cdot \ln[\int Q_C \cdot (\int P_{ch} + 0.09) \cdot \int P_{bag}]$$

P_{ret}: retainer-ring pressure (polishing mean); $\int$: time integral in chamber 4
Q_C: slurry flow C; P_{ch}: chamber pressure; P_{bag}: main air-bag pressure
(variables scaled by their median)

Table	a_g (slope)	b_g (offset)
A-456	34.5	76.6
B-456	42.3	87.0

5-fold re-discovery check

Equation searched again in each fold without its held-out 20%

Out-of-fold RMSE	Joint SR	Lin. reg.
A-456	4.55	4.53
B-456	4.80	4.94

A-456: Joint SR better in 2/5 folds

B-456: Joint SR better in 4/5 folds

Pre-defined form (retainer × log of integrals): 1/5 folds

No fold re-found the full-data equation; f values correlated with it at |r| = 0.95–0.99 (form differs).

Conclusion

The equation form was not reproducible: when re-searched in each fold, the retainer × log(integrals) form reappeared in only 1 of 5 folds, although every fold used the slurry-flow and air-bag pressure integrals. Against linear regression with the same variables, the gain was unclear for A-456 (better in 2/5 folds) and small for B-456 (better in 4/5 folds; out-of-fold RMSE 4.80 vs 4.94).

Post-hoc follow-up on Train+Val only (no test labels). Candidate variables were pre-selected on all Train+Val data, so this is not an external validation.